RESEARCH · 02

Not All Goals Are *Equal*.

A TIME-AWARE EXPECTED-POINTS MODEL · TOP 5 LEAGUES + EREDIVISIE

2026·05

renball

TL;DR — three sentences

— A goal that takes a team from 0–0 to 1–0 in the final fifteen minutes is worth 1.68 expected points — 2.1× the same transition in the opening fifteen.

— Across 39,381 goals in six leagues and seven seasons we observed 82 of 90 theoretical (gd × minute_bin) states.

— The model's biggest gainer over the time-blind baseline is Mário Rui (+3.43 ΔEP on 28 G+A) — mostly assist-type credits that land in high-value minutes.

THE QUESTION

Are all goals *equal*?

Three goals, three different lengths of game ahead. Should we treat them the same?

EARLY OPENER	0–0 to 1–0 in the 5th minute. 85 minutes still to play, everyone resets.
LATE EQUALISER	1–2 to 2–2 in the 88th minute. Pulls a point out of a lost match.
CONSOLATION	4–0 down, you score in stoppage. Doesn't move the table.

→ *The standard scoreboard treats them the same. We won't.*

DATA

What we built the framework on

39,381	6	7	82
GOALS ANALYZED	LEAGUES	SEASONS	MODELLED STATES

Seasons in scope: 2017-2018 → 2023-2024

Leagues: Bundesliga, Eredivisie, La-Liga, Ligue-1, Premier-League, Serie-A

Source: FBref-derived shot tables, one pickle per league (match_data_final_<league>.pkl).

Variables: scoring team, outcome=goal, goalscorer, SCA1 (assist), minute (parsed including stoppage suffixes).

THE VALUE FUNCTION

How a goal gets *priced*

For every pre-goal state (goal_difference, minute_bin) we compute the empirical expected points.

$$EP(gd, \text{minute_bin}) = 3 \cdot P(W \mid \text{state}) + 1 \cdot P(D \mid \text{state})$$

Mean of final-match points over every team that entered this (gd, bin) state.

A goal's value compares transitions in the same minute bin:

$$\text{ep_collected} = EP_T(g \to g{+}1, \text{bin}) - EP_T(g{-}1 \to g, \text{bin})$$

EP_T = mean final points of the teams that made that transition in that bin. The lowest observed transition in each bin is set to 0.

STATE VALUE TABLE

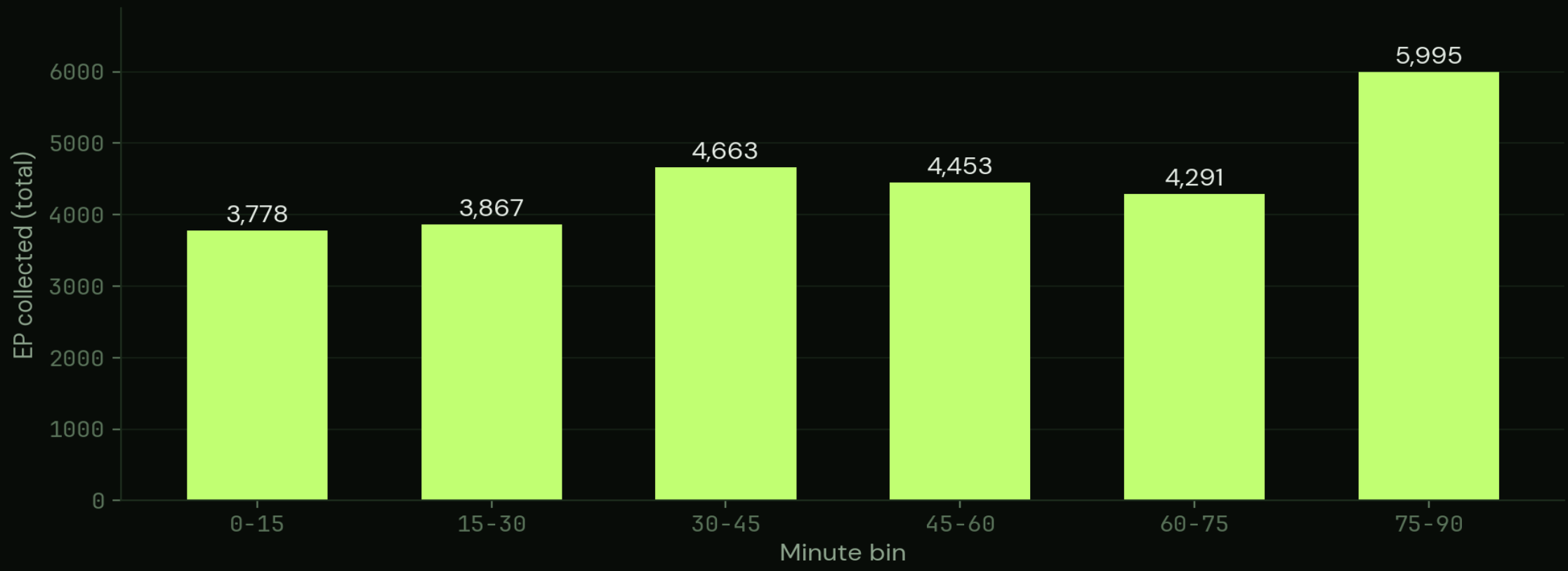
Expected final points at each (GD, minute)

	0-15	15-30	30-45	45-60	60-75	75-90
-4 TO -3		0.00	0.00	0.02	0.00	0.00
-3 TO -2	0.00	0.00	0.04	0.05	0.03	0.00
-2 TO -1	0.38	0.33	0.24	0.21	0.18	0.07
-1 TO 0	0.75	0.75	0.74	0.70	0.64	0.56
0 TO +1	1.40	1.40	1.39	1.40	1.41	1.46
+1 TO +2	2.07	2.07	2.07	2.08	2.10	2.07
+2 TO +3	2.53	2.56	2.66	2.70	2.73	2.88
+3 TO +4	3.00	3.00	2.93	2.93	2.95	3.00
+4 TO +5		3.00	3.00	2.97	3.00	3.00

Showing GD ∈ [-4, +4]; the full table covers [-7, +7]. Cells brighten with expected points.

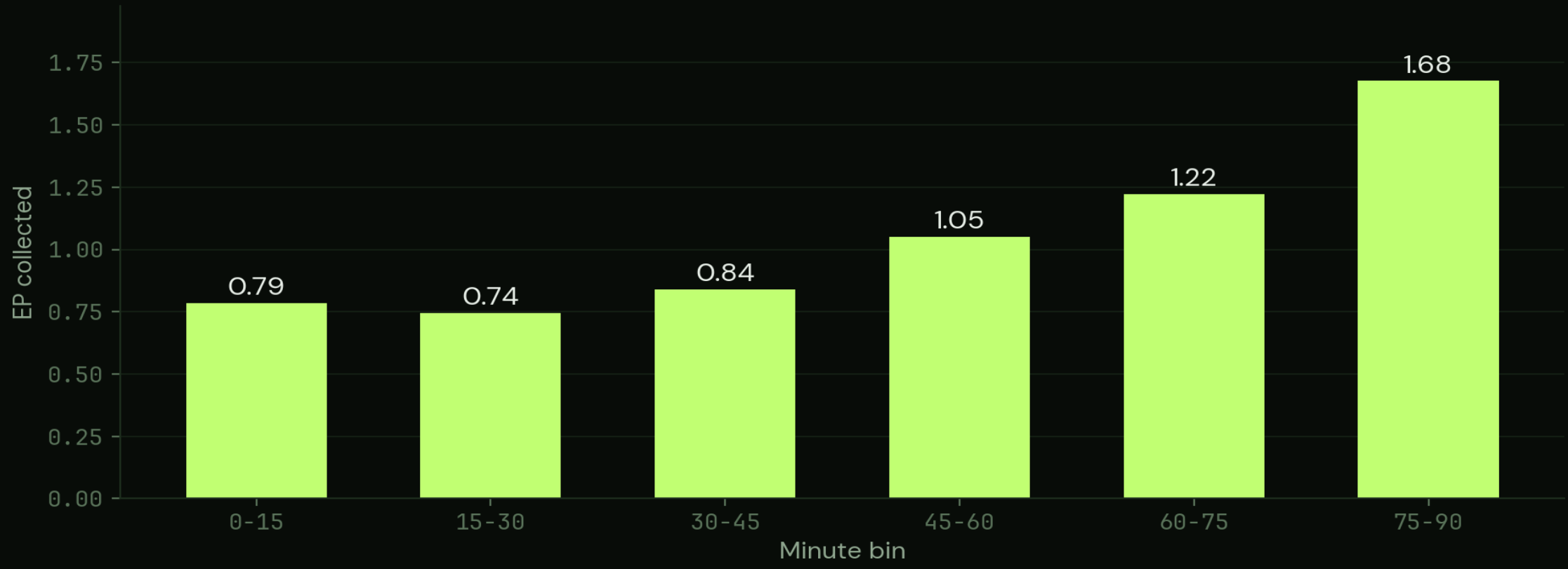
AGGREGATE

Where the expected points get *cashed in*



Sum of (ep_collected × n) across every transition, per minute bin. The last 15 minutes deliver ~1.6× the value of the first.

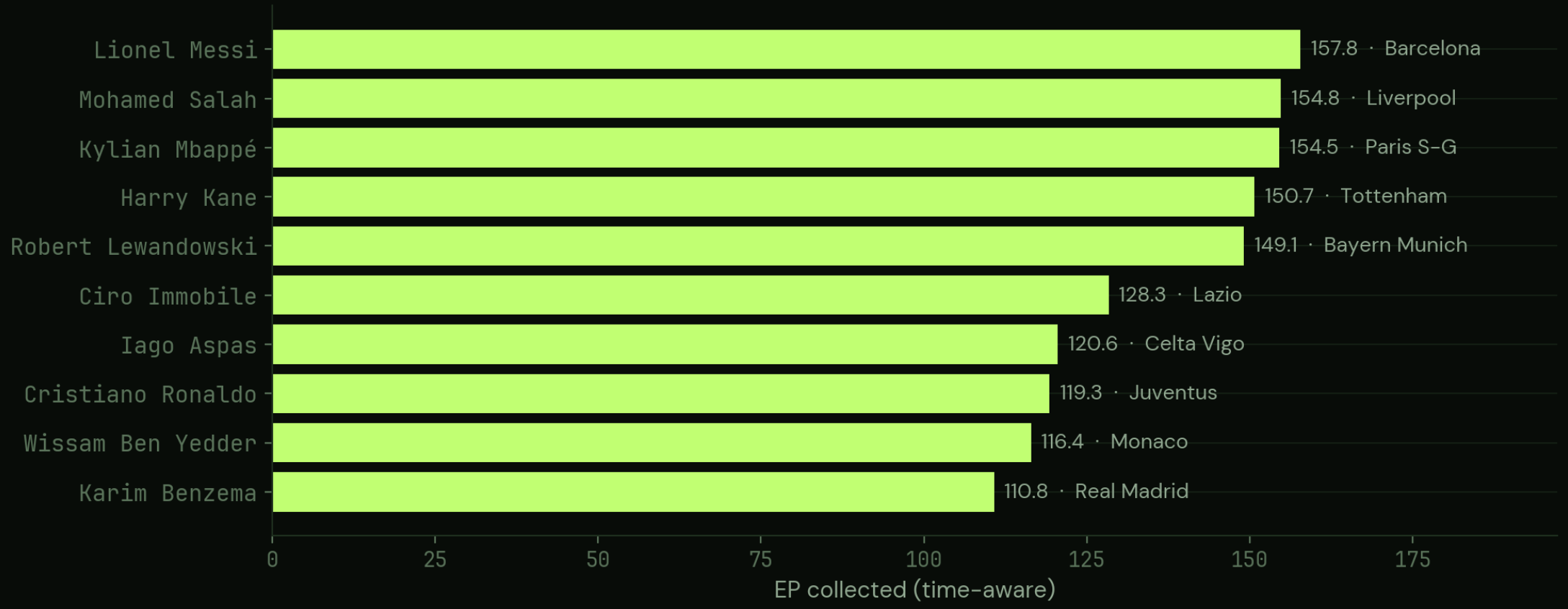
The opener: worth more the *longer it took* to arrive



EP collected from a 0→1 goal by minute bin. Late opener (1.68 EP) is 2.1× the early one (0.79 EP).

LEADERBOARD

Top 10 EP collectors · Goals + Assists



All leagues, all seasons pooled. Goals + assists combined; players pooled across team transfers (primary team labelled).

SPOTLIGHT

The biggest *time-aware* upgrade

BIGGEST GAINER

Mário Rui

Napoli · 28 G+A pooled

+3.43

Δ EP FROM TIME-AWARE RE-RATING

(24.11 timed - 20.68 untimed)

TOP 5 UPGRADES

1	**Mário Rui**	**+3.43**
2	Evann Guessand	+3.34
3	Cristian Tello	+2.82
4	Adama Traoré	+2.69
5	Youssoufa Moukoko	+2.58

Mostly assist-type credits (SCA1), not goals — landing unusually often late in the game.

IMPLICATIONS

What this changes

— Goals are not a unit — they are a function of state and time. EP-weighted rankings expose contribution patterns raw counts can't see.

— Recruitment with leverage in mind: a player whose 12 goals are mostly late equalisers is materially different from one whose 12 are 4–0 cushions.

— Squad-management: knowing which minute bins your forwards harvest from informs substitution timing and game-state-conditional roles.

— Coaching: late-minute defending decisions get a clean valuation — the equaliser you concede in the 88th minute costs far more than the one in the 18th.

LIMITATIONS

What this isn't

— Not xG-weighted. A tap-in and a 30-yard volley count the same.

— Penalties are not separated from open play, and own goals are missing from the shot data.

— Assist credit caps at SCA1 — build-up beyond the first shot-creating action isn't captured.

— Observational, not causal: it values the gap between states, not the goal's incremental effect on the final outcome.

— Pooled value function across leagues — per-league calibration available in the dashboard.

Full discussion: methodology.html · §5

Want every cell, every league, every season?

Full methodology, source code and data:

renball.com

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/PROJECTS/02-POINTS-BY-GAMESTATE/

→ analysis.html        editorial write-up with five Chart.js + D3 visuals
→ dashboard.html       full interactive Tabulator dashboard, six datasets, drill-down
→ methodology.html     formal write-up — state-value math, attribution, eight limitations
→ code/build.py        Python pipeline from raw pickles to JSON
```